

Team Project Final Report

INST123 | 0103 | Football | 12/09/2024 | Team 2

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Introduction:

Our database is centered on one of the most entertaining and highest grossing sports organizations in the world: the NFL. Data on different players and teams in the NFL in the year 2023 have been collected and organized within our database. This allows for a statistical analysis that can be used by fans and spectators to NFL football games, as well as fantasy football fans, etc. Fans who play fantasy football and place bets on players and teams can use our data in order to make more informed decisions. Using statistics like the amount of passing yards a player had or the number of touchdowns a player scored can help these fans predict the future performance of these players. Fans can leverage this data to help them set their fantasy football lineups and predict individual player outcomes to improve their chances of winning. Even passionate NFL fans who do not participate in the mentioned activities can use the statistics to fuel discussions, compare players, and stay informed on their favorite teams and players. Whether it be for competition, profit, or enjoyment, our database will allow fans to increase their engagement with the sport.

Database Description:

Physical Database

We included in our database data on player statistics for how they performed in the year 2023 and also what their physical attributes were for that year. Categories of columns include rushing yards, touchdowns, passes, fumbles, recoveries, passing yards, and more for the players. Then for teams as a whole we included wins, losses, ties, seed, cap space, head coach, staff, points scored, and points conceded. We had rows for each player and each team. For primary keys we used the `player_id` column, and then for the teams the `abbreviation` column. The team abbreviations can also be used as a foreign key to associate players in the player table with their respective team.

This all resulted in five tables:

- the `combine_data` table which included `player_name` text, `player_id` text, `position` text, `college` text, `year_drafted` integer, `team_drafted` text, `draft_round` integer, `draft_ovr` integer, `forty` numeric(5,2), `bench` integer, `vertical` numeric(5,2), `broad_jump` numeric(5,2), `cone` numeric(5,2), and `shuttle` numeric(5,2):

full_name	forty	vertical	bench_reps	broad_jump	cone	shuttle
John Abraham	4.55	NULL	NULL	NULL	NULL	NULL
Shaun Alexander	4.58	NULL	NULL	NULL	NULL	NULL
Damell Alford	5.56	25	23	94	8.48	4.98
Kyle Allamon	4.97	29	NULL	104	7.29	4.49

- the contract_data table which included player_name text, player_id text, team text, year_signed integer, length_years integer, contract_amount numeric(15,2), apy numeric(15,2), guaranteed_amount numeric(15,2):

Player Name	Team	Year Signed	Years	Contract Value	APY	Guaranteed Amo	APY as % of Cap at Signing
A. J. Brown	PHI	2024	3	96000000	32000000	84000000	12.5
A. T. Perry	NULL	NULL	NULL	NULL	NULL	NULL	NULL
Aaron Jones	MIN	2024	1	7000000	7000000	7000000	2.7
Aaron Rodgers	NYJ	2023	3	112500000	37500000	75000000	16.7
Abram Smith	MIN	2023	1	750000	750000	0	0.3
Adam Prentice	NO	2024	1	1130000	1130000	0	0.4
Adam Thielen	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- the player_physical_attributes table which included player_name text, player_id text, position text, team text, number integer, height text, weight integer, and birth_date date:

player_name	last_name	first_name	team	number	position	height	weight	birth_date	college	age	team_drafted	year_drafted
AJ McCammon	McCammon	AJ	CIN	NULL	QB	6'3"	220	1990-09-13	Alabama	34	CIN	2014
Aaron Rodgers	Rodgers	Aaron	NYJ	8	QB	6'2"	223	1983-12-02	California	40	QB	2005
Adrian Martinez	Martinez	Adrian	FA	15	QB	6'2"	220	2000-01-07	Kansas State	24	DET	2023
Aidan O'Connell	O'Connell	Aidan	LV	12	QB	6'3"	210	1998-09-01	Purdue	26	LV	2023
Andy Dalton	Dalton	Andy	CAR	14	QB	6'2"	220	1987-10-29	TCU	36	CIN	2011
Anthony Brown	Brown	Anthony	FA	NULL	QB	6'1"	217	1998-07-27	Oregon	26	BAL	2022
Anthony Brown	Brown	Anthony	FA	35	QB	5'11"	192	1993-12-15	Purdue	30	DAL	2016
Anthony Richardson	Richardson	Anthony	IND	5	QB	6'4"	244	2002-05-22	Florida	22	IND	2023

round	pick
2	112
0	16
NULL	NULL
2	92

- the ppr_player_stats_2023 table which included player_name text, player_id text, position text, team text, passing_yards integer, passing_touchdowns integer, interceptions integer, rushing_yards integer, rushing_touchdowns integer, receptions integer, receiving_yards integer, receiving_touchdowns integer, return_touchdowns integer, fumble_touchdowns integer, two_point_conversions integer, fumbles integer, touches integer, targets integer, rz_touches integer, rz_targets integer, rank integer, and total_points numeric(10,2):

player_name	team	position	passing_yards	passing_touchdowns	interceptions	rushing_attempts	rushing_yards	rushing_touchdowns	targets	receptions	receiving_yards	receiving_touchdowns
AJ McCammon	CIN	QB	19	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
Aaron Rodgers	NYJ	QB	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
Adrian Martinez	FA	QB	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
Aidan O'Connell	LV	QB	2218	12	7	17	11	1	NULL	NULL	NULL	NULL

return_touchdowns	fumble_touchdowns	2_point_conversions	fumbles	total_points
NULL	NULL	NULL	NULL	0.7
NULL	NULL	NULL	NULL	
NULL	NULL	NULL	NULL	
NULL	NULL	NULL	4	121.8

- the team_info table with abbreviation text, city text, name text, conference text, division text, head_coach text, cap_space numeric(15,2), stadium_name text, stadium_capacity integer, playing_surface text, and stadium_type text:

abbreviation	city	name	conference	division	full_name	bye_week	head_coach	offensive_coordinator	defensive_coordinator	special_teams_coach	cap_space	active_cap_spendin
SF	San Francisco	49ers	NFC	West	San Francisco 49ers	9	Kyle Shanahan	NULL	Nick Sorensen	Brian Schneider	55718800	216701619
CHI	Chicago	Bears	NFC	North	Chicago Bears	7	Matt Eberflus	Shane Waldron	Eric Washington	Richard Hightower	6769936	238374959
CIN	Cincinnati	Bengals	AFC	North	Cincinnati Bengals	12	Zac Taylor	Dan Pitcher	Lou Anarumo	Darin Simmons	9598226	247201667
BUF	Buffalo	Bills	AFC	East	Buffalo Bills	12	Sean McDermott	Joe Brady	Bobby Babich	Matthew Smiley	3307815	188881169

dead_money	offensive_scheme	defensive_scheme	primary_color	secondary_color	tertiary_color	quaternary_color	stadium_name	stadium_city	stadium_state	stadium_capacity	stadium_playing_sur	stadium_geo_lat
24488637	3WR	4-3	AA0000	B3995D	FFFFFF	NULL	Levi's Stadium	Santa Clara	CA	68500	Grass	37.403
14511642	3WR	4-3	0B162A	E64100	FFFFFF	NULL	Soldier Field	Chicago	IL	61500	Grass	41.86232
8142245	3WR	4-3	0	FB4F14	FFFFFF	NULL	Paycor Stadium	Cincinnati	OH	65515	Artificial	39.095309
63954923	3WR	4-3	00338D	C60C30	707271	FFFFFF	Highmark Stadium	Orchard Park	NY	71608	Artificial	42.77358

stadium_geo_long	stadium_type
-121.970274	Outdoor
-87.616699	Outdoor
-84.516003	Outdoor
-78.786837	Outdoor

Sample Data

We gathered realistic and accurate data about football from multiple sources across the web. Our main sources are a Github repository with plenty of CSV files over multiple years of data. We will focus mainly on the 2023 folder and only pull data from there. We supplemented this with data from SportsDataIO, Over The Cap, and Pro Football Reference for data on player's physical attributes, contract data, and their draft data. We also verified our data by cross referencing it with ESPN and the B10 Network to validate/gather more information. All in all, these things combined formed our sample dataset.

Queries

Here are the descriptions of the sample queries we used for our project database:

- Find the top 10 players with the highest total points who are quarterbacks and have a passing touchdown-to-interception ratio greater than 2.
 - Requirements Met:** Joins, filtering
- List the players and their respective team names, college, and draft details for those who were drafted in the 1st round and have more than 5 receiving touchdowns.
 - Requirements Met:** Joins, filtering

- Find the average APY (average per year value) of contracts for players by team for those who signed contracts from 2020 onwards, and show the result sorted by the highest average APY.
 - **Requirements Met:** Filtering, aggregation
- Display players who have bench reps greater than the average bench reps of all players, along with their position, team, and combine metrics.
 - **Requirements Met:** Joins, filtering, subquery
- Count the number of players per team who had more than 250 total points and played in positions other than quarterback.
 - **Requirements Met:** Joins, filtering, aggregation

	Requirement A JOIN	Requirement B FILTER	Requirement C AGGREGATE	Requirement D SUBQUERY
Query 1:	X	X		
Query 2:	X	X		
Query 3:		X	X	
Query 4:	X	X		X
Query 5:	X	X	X	

Our implementation of these queries can be found in the project deliverable(queries.txt)

Changes from original design:

We made a few changes to the database design since the last Progress Report, since the available data was not as complete as we had hoped. We removed hand size and arm size from the player_physical_attributes table, as well as apy_pct_of_cap from the contract_data table, replacing it with guaranteed_amount.

Lessons Learned:

This project gave us real experience using the concepts and knowledge of SQL that we built throughout this class. Being able to take what we learned and apply it to a real database has made our understanding much more concrete. It has also helped clarify the potential use cases of some features of SQL that weren't as clear when learning about them from a lecture. As the

culmination of everything that is covered in this class, this project improved our understanding of how each concept from the class fit together as opposed to taking place in atomic parts.

For our project specifically, we had to narrow down the scope of our database by removing some columns. This meant that we gained experience in thinking more carefully about what is really needed to answer the questions that we wanted to answer. In the real world, gathering additional data can represent extra costs to a project, so it's good to have a strong grasp of what data is truly required before setting out on the data collection portion of a project.

A lot of the data we used was a little messy. There were some duplicate records and things that we had to clean up. This was a good reminder of the nature of real-world data, which tends to have errors due to the natural messiness of reality. We did our best to clean up what we could so that any analysis we did could be as accurate as possible, while also representing a complete picture of the domain that we are working in.

Potential Future Work:

One limitation for our database was that it only had a scope for the year 2023. So one area for potential future work is to include data from other years. This way, the database can be used to track trends over time. It could also be interesting to attempt to generate statistical models based on the data using functions like `regr_slope` and `regr_intercept`. Using the results of these two pieces of work, there could also be interesting visualizations which could be made to help people engage with and learn from this data.

Team Member Participation:

Ryan Combs, Akhil Karumuri, Connor Criss, and Eric Liu contributed to the final project deliverables.

Citations:

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